

in their software development and testing cycle. This infrastructure is also developing very fast in cloud technology and cloud based services.

History of Linux containers

Container technology has evolved from a number of previous innovations and ideas. Starting from 1979, a lot of work has been done in the field of containerisation, which has given birth to present day container management systems like Docker.

■ 1979 - UNIX V7

When the development of UNIX V7 was under way in 1979, the chroot system call was invented for changing the root directory of a process and its children to a new file system location. The thinking behind this was to provide an isolated hard drive space for each process. This was the first step in the field of containerisation.

■ 2000 - FreeBSD Jails

Approximately 20 years later, the owner of R&D Associates Inc., Derrick T. Woolworth, came up with the earliest container technology in 2000 and named it FreeBSD Jails. This was done in order to partition a computer system into numerous independent smaller systems called jails. It had the ability to assign an IP address, custom software installations and configurations.

■ 2001 - Linux VServer

This was similar to FreeBSD's jail mechanism and could be used to securely partition resources on a computer file system by patching a Linux kernel. Each partition is given the name and security context, and the virtualised system within is called a virtual private server. Its last stable patch was released in 2006.

■ 2004 - Oracle Solaris container

Oracle released the beta version of the Solaris container in 2004 and the full release took place in 2005, which was a combination of system resource controls and boundary separation provided by zones. These zones act as a complete isolated server within a single operating system instance.

■ 2005 - OpenVZ (Open Virtuozzo)

This is similar to the Solaris container in its implementation of a patched Linux kernel for isolation, resource management and virtualisation purposes. Each container has an isolated file system, its users and user groups, devices, networks, etc.

■ 2006 - Process containers

Process containers were introduced by Google in 2006 to limit and isolate resource usage of a collection of processes. These were later renamed Control Groups in order to avoid being confused with the 'container' used in the Linux kernel context, and were ultimately merged with Linux kernel 2.6.24.

■ 2008 - LXC

LXC was the first and the most complete Linux container. It was developed by using cgroups and Linux namespaces. It did not require any patches to work and was made to

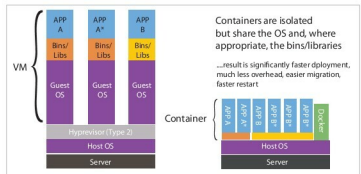


Figure 1: Differences between the architecture of VMs and containers.

Image credit: Docker Inc.

work on the vanilla Linux kernel. It is widely used since it provides support for different languages such as Java, Ruby, Python3, etc.

■ 2011 - Warden

CloudFoundry developed Warden in 2011. This container was not limited to the Linux operating system; instead, it was made to work on any OS. Though in the initial stages of its development it was based on LXC, it later replaced LXC with its own execution. It works by isolating environments by running on daemons, and providing an interface for managing the containers.

■ 2013 - LMCTFY

LMCTFY or 'Let me contain that for you', can be thought of as an open source form of Google's container stack. The aim behind this project was to develop Linux application containers with high utilisation of shared resources, so as to get the maximum performance with the containers. This project proved to be a benchmark in the field, since many organisations have built tools as a result of it. After Google's contribution of the central LMCTFY ideas to libcontainer, the work on this project stopped in 2015.

■ 2013 - Docker

When Docker appeared in 2013, it was the most efficient container management system and is still rated as the market leader. It was initially named dotCloud but was later renamed to Docker. It uses its own container library known as libcontainer for its management, though initially it made use of LXC just like Warden. It differed from the earlier container management systems in the way that it was a complete ecosystem for handling containers.

■ 2016 - Windows containers

After watching the popularity of containers in the Linux operating system, Microsoft released Microsoft Windows Server 2016.

Security of containers

The security of Linux containers is of paramount importance, especially if you are dealing with sensitive data like in the banking domain. Since different software is installed on different containers, it becomes very important to secure your container properly to avoid any